

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | Bengaluru, India Jul 2023 – Jul 2025

- Owned a multi-tenant NHS eTriage backend (Node.js, TypeScript, MongoDB) shipped to production for **8+ UK clients**, processing **10,000+ monthly submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Built a full-stack data annotation platform (React front-end, Node.js/MongoDB back-end) that turned messy manual labeling into structured training data for the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Ran a COO-sponsored GitHub Copilot pilot, routing tech-debt PRs, code migrations, and fix triage through AI-assisted workflows and maintaining a versioned conventions file to keep AI-generated code aligned with team patterns.
- Implemented production security and reliability controls, AWS KMS encryption for sensitive data, auth hardening, rate limiting, and integration tests, remediating **20+ VAPT findings**; served as primary technical POC for the stack.

Cario Growth Services, Backend Developer Intern | Chennai, India Nov 2022 – May 2023

- Migrated core repositories from JavaScript to **TypeScript** and shipped a production LLM auto-commenting feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation, per-persona prompts, and content-safety filtering.
- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.

Freelance / Open Source, Full-Stack Engineer | Remote Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN web platform (React front-end) with Stripe hybrid payments, cron automation, and multi-branch support; live **3+ years**.
- Delivered a second MERN client engagement (US-based inventory management system) end-to-end as a paid freelance build.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic service that turns open-ended queries into structured entity tables from unstructured web data via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research assistant in **TypeScript** on Cloudflare Workers: an agent that fetches web sources, generates multi-step research digests, and streams responses over WebSocket with durable, restart-safe workflows.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline that turns documents into structured, retrievable knowledge (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: TypeScript, JavaScript, Python, SQL, C++, Java, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: React, Node.js, Express, FastAPI, Cloudflare Workers, REST APIs, Docker, Git, GitHub Actions

ML & AI: LLM agents, RAG, structured extraction, OpenAI / Groq APIs, PyTorch, Hugging Face Transformers, evaluation

Retrieval / Search: RRF, cross-encoder reranking, MMR, sentence-transformers, cell-level provenance, Brave Search

Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.